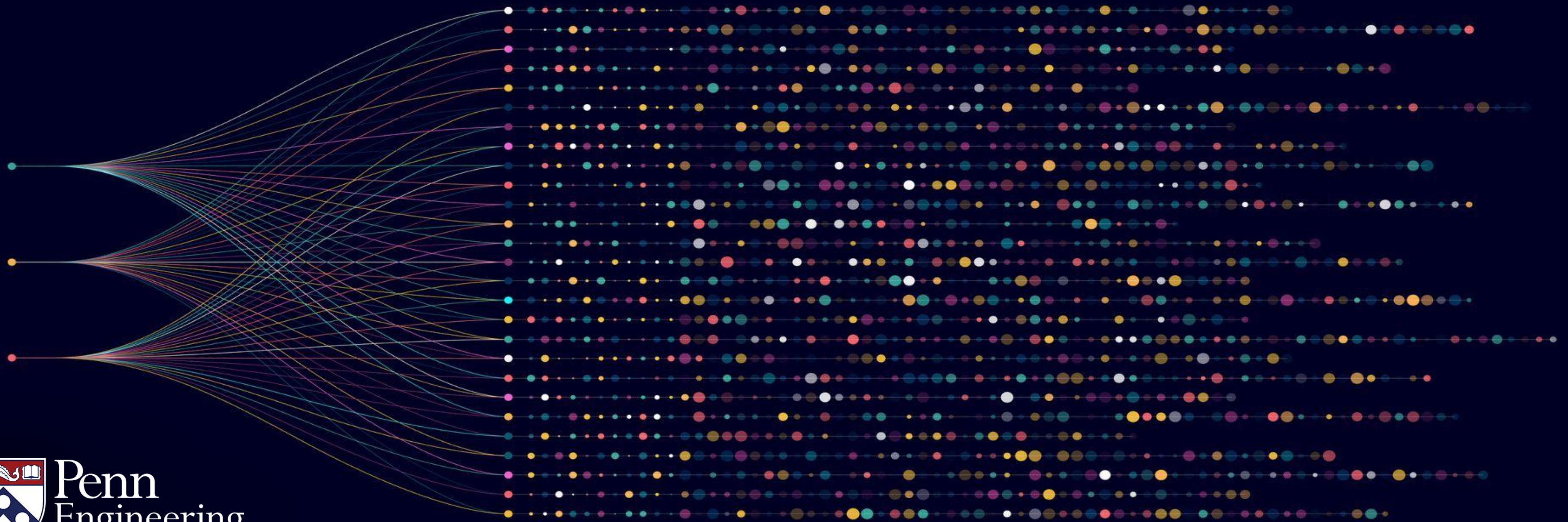


Algorithms for Big Data

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Independence of random variables



An old saying

- Old saying: “Lightning never strikes the same place twice”
- Clearly False: There are lightning prone areas



Independent events?

- Lightning affected by temperature, moisture, altitude etc. -- so areas prone to lightning often remain so for extended durations



Independence of factors?

Implicitly, this saying assumes that lightning strikes are **independent events** – an assumption which is clearly false!

Arguably, independence of events is the most (mis)used assumption – both in theory and practice

Examples: 2008 financial crisis – mathematical models largely ignored correlations between investors

What is independence?

Formally, two random variables $\mathbf{A}$ and $\mathbf{B}$ are said to be independent if for every possible choice of x and y , we have that:

$$\Pr[\mathbf{A} = x \text{ and } \mathbf{B} = y] = \Pr[\mathbf{A} = x] \cdot \Pr[\mathbf{B} = y].$$

More generally, a set of random variables $\mathbf{A}_1, \dots, \mathbf{A}_n$ are mutually independent if for every choice of $x_1, \dots, x_n$, it holds that

$$\Pr[\mathbf{A}_1 = x_1, \mathbf{A}_2 = x_2, \dots, \mathbf{A}_n = x_n] = \prod_{i=1}^n \Pr[\mathbf{A}_i = x_i].$$

Why independence?

Succinct representation: If the variables $A_1, \dots, A_n$ are mutually independent, then the joint distribution of $A_1, \dots, A_n$ can be described using just the individual random variables $A_1, \dots, A_n$.

Example: Suppose $A_1, \dots, A_n$ are Bernoulli random variables, i.e., take values in $\{0,1\}$.

Useful notation: $\{0,1\}^n$ denotes the set of n –tuples whose coordinates comes from $\{0,1\}$.

(For example, for $n = 2$, $\{0,1\}^2 = \{(0,0), (0,1), (1,0), (1,1)\}$)

Note that $(A_1, \dots, A_n)$ takes values in $\{0,1\}^n$.

Exponential description of joint probability distributions

If $A_1, \dots, A_n$ are **not independent**, then in the worst case, specifying the joint distribution of $A_1, \dots, A_n$ would require specifying $\Pr[A_1 = x_1, A_2 = x_2, \dots, A_n = x_n]$ at every point $(x_1, \dots, x_n)$

In the worst case, that will take specifying $\Theta(2^n)$ real numbers

** For those who are not familiar with the big Theta notation, $f(n) = \Theta(g(n))$ is equivalent to $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.*

Succinct size for independent random variables

Suppose $A_1, \dots, A_n$ are **independent** Bernoulli random variables such that for each $1 \leq i \leq n$, $\Pr[A_i = 1] = \rho_i$ and $\Pr[A_i = 0] = 1 - \rho_i$. Succinctly, **$\Pr[A_i = x_i] = 1 - \rho_i + x_i \cdot (2\rho_i - 1)$** .

$(A_1, \dots, A_n)$ can take values $(x_1, \dots, x_n) \in \{0,1\}^n$ and

$$\Pr[A_1 = x_1, A_2 = x_2, \dots, A_n = x_n] = \prod_{i=1}^n \Pr[A_i = x_i] = \prod_{i=1}^n (1 - \rho_i + x_i \cdot (2\rho_i - 1)).$$

Summary: If $A_1, \dots, A_n$ are **independent**, then the joint distribution of $(A_1, \dots, A_n)$ can be specified using n real numbers $\rho_1, \dots, \rho_n$.

Mathematically tractable behavior

Mathematically tractable behavior: Independent random variables are easy to analyze – many of the most fundamental results in probability such as the Central limit theorem or Chernoff bounds require independence

Theorems in probability theory which require no independence exist but are quite rare

What makes random variables independent?

Physical independence of underlying actions: If the physical action governing the random variables are **independent**, then the random variables are independent

Example: If we toss 2 coins *independently*, then the outcome of the first coin (H, T) is independent of the outcome of the second coin (H, T)

What makes random variables independent?

Subtle examples of independence: Sometimes random variables can be independent even when the underlying physical actions are quite dependent

Example: roll two fair six-sided dice independently

$X = 1$ if the first die is (2, 4, 6) and $X = 0$ otherwise

$Y = 1$ if the sum of the dice is 7 and $Y = 0$ otherwise

Seemingly dependent (because the sum of the dice is dependent on the outcome of the first)
but turns out that X and Y are independent **Verify: left as an exercise**

Summary

- There are many other "subtle" examples of independence – e.g., in the context of continuous random variables, stochastic processes
- See some examples later in the course
- Independence of random variables important in both theory and practice